ModelIt!

Teacher's Guide

We Built a Better Dog (And Maybe a Mistake)

How Humans Hack Evolution Through Selective Breeding

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-LS4-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS4-5

Gather and synthesize information about technologies that have changed the way humans influence the inheritance of desired traits in organisms.

Three-Dimensional Learning

Science Practice	**Obtaining, Evaluating, and Communicating Information** — Students gather and synthesize information about how technologies like selective breeding and genetic modification change inheritance of traits.
Disciplinary Core Idea	**LS4.B Natural Selection and Artificial Selection** — In artificial selection, humans have the capacity to influence certain characteristics of organisms by selective breeding.
Crosscutting Concept	**Cause and Effect** — Students identify that human selection pressure CAUSES changes in trait frequency, with both intended effects (desired traits) and unintended consequences (reduced diversity).

CAST Testing Alignment

MULTIPLE CHOICE:

Over 15,000 years, humans selectively bred wolves into hundreds of dog breeds. Which statement best explains why all dog breeds can still interbreed with wolves?

CONSTRUCTED RESPONSE:

Explain how selective breeding can both improve a desired trait AND create unintended health problems. Use a specific example in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Humans have been modifying organisms through selective breeding for over 10,000 years — long before we understood genetics. From wolves to dogs, from wild grass to corn, artificial selection has shaped the living world to meet human needs.

From Wolves to Dogs

All domestic dogs (Canis lupus familiaris) descended from wolves through at least 15,000 years of selective breeding. Early humans selected wolves that were less aggressive and more cooperative. Over centuries, specialized breeding created everything from tiny Chihuahuas to massive Great Danes — all from the same ancestor.

Artificial vs. Natural Selection

Both processes work through the same mechanism: variation exists, some individuals reproduce more than others, traits are inherited. The difference is WHO selects. In natural selection, the environment determines fitness. In artificial selection, HUMANS choose which traits to favor — often for our benefit rather than the organism's.

The Cost of Extreme Breeding

Extreme selective breeding reduces genetic diversity, causing problems. Purebred English Bulldogs have such flat faces they can barely breathe. German Shepherds have sloped backs causing hip dysplasia. Cavalier King Charles Spaniels often have skulls too small for their brains. These are direct consequences of breeding for appearance over health.

Modern Genetic Technologies

Today, selective breeding has been supplemented by genetic modification (GMOs), gene editing (CRISPR), and cloning. These technologies can achieve in one generation what would take decades of selective breeding. They raise important ethical questions about how far humans should go in modifying other organisms.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify selection pressure, generation time, desired trait frequency, and genetic diversity as components of the selective breeding system.

E **Establish Relationships**
Students determine that selection pressure increases desired traits but decreases overall genetic diversity — revealing the trade-off.

V **Visualize & Model**
Students build a model showing how human selection changes populations over generations.

E **Evaluate Outcomes**
Students run mild, extreme, and no-selection scenarios to compare evolutionary outcomes.

R **Revise & Extend**
Students add inbreeding depression or GMO technology to explore more complex breeding dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every dog breed on Earth came from ONE ancestor: the wolf. We did that. But should we have?"*

Watch For

Show wolf → dog breed comparison. Ask: How did we get from wolf to Chihuahua?

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating how humans hack evolution — and the consequences."*

Watch For

Have students read objectives. Pre-teach 'artificial selection.'

Slide 3: Big Question

3 min

Say: *"Corn used to be a tiny grass. Dogs used to be wolves. How did WE do that?"*

Watch For

Show before/after of corn (teosinte → modern corn). Discuss.

Slide 4: LEVER Framework

2 min

Say: *"We'll model how selection pressure changes populations over generations."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What do humans control in selective breeding? What changes as a result?"*

Watch For

Guide sorting. Discuss why selection pressure is human-controlled.

Slide 6: Activity 2: Connections

8 min

Say: *"Stronger selection gets you the trait faster. But what's the hidden cost?"*

Watch For

Reveal the negative relationship to genetic diversity.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare: gentle selection vs. extreme purebred breeding. What's the trade-off?"*

Watch For

Students predict first. Show the diversity crash in extreme selection.

Slide 8: Discoveries

5 min

Say: *"Why can't English Bulldogs breathe properly? What did our model reveal?"*

Watch For

Show bulldog health problems. Connect to model's diversity crash.

Slide 9: STEM Challenge

12 min

Say: *"Design a breeding program that gets the trait you want WITHOUT the cost."*

Watch For

Groups design programs balancing trait selection with diversity.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Selection Pressure, Generation Time
- **INTERNAL:** Desired Trait Frequency, Genetic Diversity

Relationships

- **Selection Pressure to Desired Trait Frequency:** POSITIVE (+) — Stronger selection pressure means only organisms with the desired trait reproduce, rapidly increasing how common that trait is in the population.
- **Generation Time to Desired Trait Frequency:** POSITIVE (+) — More generations of selection allow the desired trait to become more common. Faster-reproducing organisms show results sooner.
- **Selection Pressure to Genetic Diversity:** NEGATIVE (-) — Strong selection eliminates individuals without the desired trait, reducing the gene pool. Extreme selection dramatically narrows genetic diversity, making populations vulnerable.

Key Discoveries

- Artificial selection works the same as natural selection, but humans choose the 'fittest' traits instead of nature
- Extreme selection increases desired traits but DECREASES genetic diversity — making populations vulnerable
- Many purebred dogs suffer health problems directly caused by selective breeding for appearance
- All domestic dogs evolved from wolves through thousands of years of human-directed selection

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Selective breeding is the same as genetic modification"	Selective breeding uses natural reproduction — choosing which organisms mate based on their traits. Genetic modification directly alters DNA in a lab. Selective breeding has been done for 10,000+ years; GMOs are a modern technology. Both change organisms, but through very different methods.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Artificial selection is unnatural"	Humans have been selectively breeding organisms for over 10,000 years — longer than recorded history. Every fruit, vegetable, and domestic animal has been modified by humans. The 'natural' versions of corn, bananas, and watermelon look nothing like what we eat today.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More selection is always better"	Extreme selection creates a dangerous trade-off. While the desired trait increases, genetic diversity crashes. Low diversity makes populations vulnerable to disease epidemics, environmental changes, and inherited disorders. The Irish Potato Famine killed over a million people because all the potatoes were genetically identical.